PREG ASSIST

Project Id
25-26J-253

Project Proposal Report

Perera A.N.M

IT21215438

B.Sc. (Hons) Degree in Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

Privacy-Preserving Real-Time Emotion Detection and Rsk Prediction for Maternal Mental Health

25-26J-253

Project Proposal Report

Perera A.N.M

IT21215438

Supervised by – Dr Mahima Weerasinghe

Co – supervisor – Ms.Osuri Duniwila

B.Sc. (Hons) Degree in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Perera A.N.M	IT21215438	

Supervisor's Certification:

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor: Date: 29/8/2025

Abstract

Pregnancy is a critical period during which maternal mental health significantly impacts both the mother and the developing fetus. Early detection and intervention for emotional distress can prevent adverse outcomes, yet existing digital mental health tools often compromise privacy or lack real-time multimodal assessment. This research proposes a privacy-preserving, generative AI mental health chatbot specifically designed for pregnant women, integrating multimodal emotion detection and personalized risk assessment. The system captures real-time facial expressions, voice prosody, and textual inputs, processing them on-device to generate a risk score across five emotional states: Happy, Sad, Stress, Anxiety, and Depressed. Raw images and audio are immediately discarded after extracting facial embeddings and acoustic features, which are securely encrypted using hardware-backed keys (Android Keystore), ensuring privacy-by-design. The generative AI component enables adaptive, context-aware conversational support, providing personalized recommendations and guidance. A fusion algorithm combines multimodal cues with chatbot interactions to produce actionable mental health insights. When risk scores exceed predefined thresholds, the system can alert a trusted contact via SMS to facilitate timely intervention. Edge models continuously improve through federated learning, allowing adaptation to diverse emotional patterns without transmitting sensitive raw data. Anticipated outcomes include high accuracy in real-time emotion recognition, effective personalized interventions, and enhanced privacy protection compared to conventional cloud-based systems. This approach demonstrates a novel integration of generative AI, on-device affective computing, multimodal fusion, and secure federated learning in maternal mental health, providing a framework for responsible and adaptive digital health tools.

Keywords: maternal mental health, generative AI, multimodal emotion detection, privacy-preserving AI, federated learning, chatbot

Table of Contents

Contents

Declaration iii
Abstract iv
Table of Contents v
1. INTRODUCTION vii
1.1 Background & Literature Survey vii
1.2 Research Gap viii
1.3 Research Problem ix
2. OBJECTIVES xi
2.1 Main Objective xi
3. METHODOLOGY xiv
3.1 System Design and Architecture xiv
3.2 Data Collection xv
3.3 Multimodal Emotion Detection xv
3.4 Multimodal Fusion and Risk Scoring xvi
3.5 Generative AI Chatbot Guidance xvi
3.6 Privacy and Security xvi
3.7 Emergency Alert Mechanism xvi
3.8 System Evaluation xvi
4.Project Requirements xviii
4.1 Functional Requirements xviii
6. BUDGET AND BUDGET JUSTIFICATION xx
7.Failure Mode Analysis xxi
8. Gantt Chart xxiii
REFERENCES xxv

1. INTRODUCTION

1.1 Background & Literature Survey

Maternal mental health is a critical factor during pregnancy, as emotional well-being directly influences both maternal and fetal outcomes. Early detection of stress, anxiety, and depressive symptoms can prevent adverse health consequences, yet conventional monitoring methods are often reactive, intermittent, or reliant on healthcare visits. Recent research has explored technological interventions to monitor and support maternal mental health, including emotion recognition systems, chatbots, and wearable sensors.

Smith, Lee, and Patel developed a mobile-based emotion recognition system using facial expressions to detect stress and anxiety in pregnant women [1]. While effective, their reliance on cloud processing for emotion analysis posed privacy risks because sensitive facial images were transmitted externally. Lee and Kim proposed a multimodal system combining voice and text inputs to predict maternal stress levels [2]. Although the approach improved predictive accuracy, it did not incorporate real-time facial analysis or generate adaptive recommendations, limiting its responsiveness in dynamic scenarios.

Chen, Zhao, and Wong implemented an AI-based chatbot providing mental health support through conversational guidance based on text input [3]. The system facilitated engagement but lacked multimodal emotion detection and real-time risk scoring, which are essential for early intervention. Wang, Zhang, and Singh employed wearable sensor-based approaches to monitor physiological signals for anxiety prediction [4]. Despite high accuracy, the requirement for additional hardware reduced accessibility and general usability.

Gupta, Sharma, and Rao demonstrated the potential of federated learning for privacy-preserving mental health prediction [5]; however, their system did not integrate on-device real-time affect detection or generative AI-driven adaptive guidance. Roberts and Thompson explored facial micro-expression recognition but targeted general populations without pregnancy-specific adaptation [6], reducing clinical relevance. Martinez, Lopez, and Fernandez designed a chatbot for pregnant women, yet it relied on cloud-based emotion analysis and lacked mechanisms to alert trusted contacts in high-risk situations [7].

These studies highlight advancements in maternal mental health monitoring but reveal a persistent gap in privacy-preserving, real-time, multimodal emotion recognition integrated with adaptive generative AI-driven guidance and emergency alert

mechanisms. The present research addresses these gaps by proposing a comprehensive AI system that captures facial expressions, voice prosody, and textual inputs, processes them on-device using edge AI, and fuses multimodal cues through a confidence-weighted fusion algorithm to compute a risk score across five emotional states: Happy, Sad, Stress, Anxiety, and Depressed. To ensure privacy, raw images and audio are immediately discarded after extracting embeddings, which are securely encrypted using hardware-backed keys (Android Keystore/iOS Secure Enclave). A generative AI chatbot provides context-aware personalized guidance, including breathing exercises, meditation, and calming activities, while federated learning allows continuous improvement of edge models without exposing sensitive user data. Additionally, the system can notify user-nominated trusted contacts via SMS when critical risk levels are detected, enabling timely intervention and proactive support for pregnant women.

1.2 Research Gap

Despite significant advancements in maternal mental health monitoring through digital technologies, several critical limitations persist, particularly in systems targeting pregnant women. First, most existing emotion recognition systems rely on cloud-based processing of sensitive facial images and audio [1][7]. This introduces serious privacy concerns, as raw data are transmitted and stored externally, which is particularly sensitive in the context of pregnancy. There is a clear need for on-device, privacy-preserving solutions that process data locally and immediately discard raw inputs after feature extraction.

Second, many systems employ single-modal analysis, such as facial expressions [1][6] or text-based chatbots [3], or combine only two modalities like voice and text [2][16]. This limits the accuracy and robustness of mental health assessments. Pregnant women's emotional states are complex and dynamic, requiring multimodal analysis that fuses facial expressions, voice prosody, and textual inputs to generate a more reliable and comprehensive risk score.

Third, most AI chatbots in maternal mental health provide predefined conversational guidance without context-awareness or adaptive personalization [3][7]. They do not leverage generative AI to deliver dynamic, situation-aware recommendations tailored to the individual's real-time emotional state. This reduces the effectiveness of interventions, especially in high-risk cases.

Fourth, although federated learning has been explored for privacy-preserving health predictions [5], recent studies have started applying it to emotion recognition. For instance, Gül *et al.* proposed a multimodal federated learning model for real-time

emotion recognition [17], while Gupta *et al.* investigated brain-signal-based emotion recognition with federated learning [18]. Similarly, Chen *et al.* introduced AFLEMP, an attention-based federated model using multimodal physiological data [19]. However, these approaches have not yet been integrated with real-time on-device emotion detection, leaving a gap in adaptive learning while maintaining user privacy. Existing systems either compromise privacy by sending raw data to the cloud or fail to continuously improve their predictive capabilities in a real-world setting.

Finally, current maternal health systems lack emergency alert mechanisms. While Martinez et al. [7] provide emotional guidance, they cannot notify trusted contacts during crises. Recent works propose IoT-based safety devices for pregnant women [21] and AI-driven maternal health interventions [20], but these remain separate from emotion-monitoring chatbots. Thus, a gap exists in integrating real-time emotional monitoring with automated emergency alerts for pregnant women.In summary, the current body of work shows gaps in privacy-preserving on-device processing, multimodal emotion detection, adaptive generative AI guidance, federated learning integration, and emergency alert mechanisms. Addressing these gaps is essential to develop a secure, real-time, adaptive, and actionable maternal mental health monitoring system, which is the primary objective of this research.

1.3 Research Problem

Pregnancy is a critical period where maternal mental health significantly affects both the mother and the developing fetus. Emotional distress, including stress, anxiety, and depression, can lead to adverse health outcomes if not detected and addressed promptly. Current digital solutions for maternal mental health monitoring face several limitations. Many systems rely on cloud-based processing, which compromises privacy by transmitting sensitive facial images, voice recordings, and textual data. Single-modal or limited multimodal approaches reduce the accuracy and reliability of mental health assessments, failing to capture the complexity of a pregnant woman's emotional state. Additionally, most existing chatbots provide static or predefined guidance without adapting to real-time emotional cues, limiting the effectiveness of interventions. Systems that implement federated learning often do not integrate it with on-device real-time processing, reducing the potential for continuous model improvement while maintaining user privacy. Furthermore, current solutions lack mechanisms for timely emergency alerts to trusted contacts in critical situations, leaving high-risk users without immediate support.

The core research problem, therefore, is the need for a secure, real-time, adaptive, and privacy-preserving maternal mental health monitoring system that can accurately detect

emotional states using multiple modalities, provide personalized and context-aware recommendations through a generative AI chatbot, and deliver timely alerts to trusted contacts when high-risk conditions are detected. Developing such a system requires integrating on-device edge AI, multimodal fusion algorithms, generative AI guidance, hardware-backed encryption, and federated learning, while ensuring usability, accessibility, and safety for pregnant women. This research aims to address this comprehensive problem, bridging gaps in accuracy, privacy, adaptability, and emergency response in maternal mental health monitoring.

2. OBJECTIVES

2.1 Main Objective

The main objective of this research is to develop a mobile application integrated with a real-time AI-based mental health monitoring system for pregnant women, which leverages multimodal inputs and generative AI to provide personalized mental health support. In today's world, maternal mental health issues such as stress, anxiety, and depression are increasingly prevalent and can have serious consequences for both mothers and their unborn children. Pregnant women often lack accessible, private, and adaptive digital tools that can monitor emotional well-being in real time and provide actionable interventions. Recognizing this, our research aims to develop a solution that continuously tracks emotional states using facial expressions, voice prosody, and text inputs, processes the data securely on-device, and computes a mental health risk score.

By integrating a generative AI chatbot, the system will provide personalized recommendations such as breathing exercises, meditation, and calming activities, tailored to the individual's emotional state. Additionally, the system incorporates hardware-backed encryption to preserve user privacy and can notify trusted contacts if a high-risk emotional state is detected, enabling timely support. The mobile application thus serves as a comprehensive, secure, and adaptive tool for monitoring and improving maternal mental health during pregnancy, combining state-of-the-art AI techniques with real-world usability and safety considerations.

2.2 Specific Objectives

1. **Develop a mobile application for pregnant women**

   Create a user-friendly and accessible mobile application that allows pregnant women to interact seamlessly with the mental health monitoring system. The app will serve as the primary platform for capturing inputs, providing guidance, and displaying mental health scores in real time.

2. **Implement multimodal emotion detection**

   Capture and analyze multiple forms of user data, including facial expressions through the camera, voice prosody from spoken inputs, and textual responses in the chatbot. This approach ensures a more accurate and comprehensive assessment of emotional states, including Happy, Sad, Stress, Anxiety, and Depressed.

3. **Integrate on-device edge AI processing for privacy**

   Process all facial images and audio locally on the user's device to extract embeddings and immediately discard raw data. Employ hardware-backed encryption (Android Keystore) to store only anonymized, secure representations of emotional data, ensuring privacy by design.

4. **Design and implement a generative AI chatbot**

   Develop a conversational AI chatbot capable of adaptive and personalized recommendations based on the real-time emotional state of the user. The chatbot will provide practical interventions, including breathing exercises, meditation, and calming activities, tailored to the severity of detected stress or anxiety.

5. **Develop a multimodal fusion algorithm**

   Create an algorithm that intelligently combines inputs from facial expressions, voice, and text to compute a composite risk score. The fusion will account for confidence levels from each modality, resulting in a reliable and holistic assessment of the user's mental state.

6. **Incorporate federated learning techniques**

   Implement federated learning to improve model accuracy and adaptability over time without transmitting sensitive raw data to central servers. This allows continuous enhancement of emotion detection models while preserving user privacy.

7. **Implement emergency alert mechanisms**

   Enable the system to notify trusted contacts (e.g., family members) via SMS when high-risk emotional states are detected. This ensures timely support and intervention, providing safety for both the mother and the unborn child.

8. **Ensure usability and accessibility**

   Design the system to be practical, easy to use, and accessible for pregnant women of different ages and technological backgrounds. The interface will minimize complexity while ensuring that all features, including real-time monitoring and guidance, are straightforward to use.

9. **Evaluate the system's effectiveness**

   Conduct systematic testing to measure the system's performance, including the accuracy of emotion detection, the reliability of risk scoring, the quality of personalized recommendations, and overall user satisfaction. Evaluation will validate the system's potential impact on improving maternal mental health outcomes.

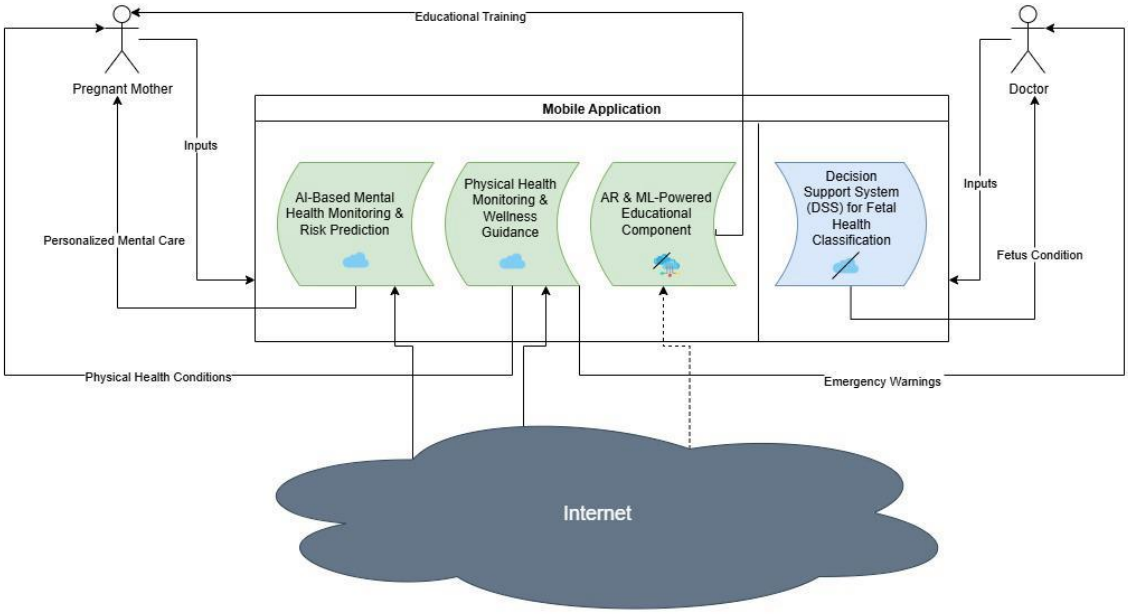
3. METHODOLOGY

The methodology of this research is designed to develop a secure, real-time, multimodal AI system for maternal mental health monitoring and personalized intervention. The system integrates facial emotion detection, text-based chatbot interaction, generative AI guidance, and federated learning to provide a comprehensive solution. The methodology is divided into the following key stages:

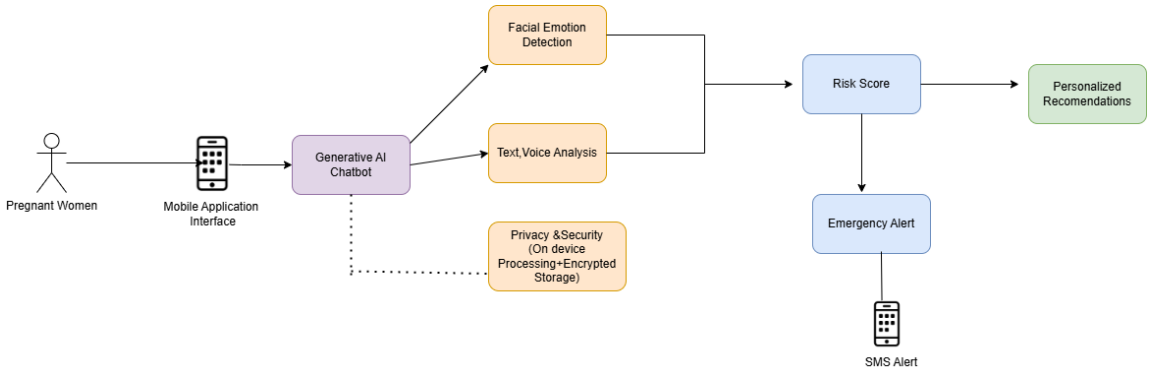
3.1 System Design and Architecture

The system consists of a mobile application interface for pregnant women and a back-end AI framework for emotion processing and guidance generation. The mobile app captures facial expressions through the camera and text input through the chatbot interface. All data are processed locally on the device using edge AI models, ensuring privacy and minimizing latency. The architecture also includes mechanisms for federated learning and trusted contact alerts.

The overall system diagram is illustrated in Figure 3.1.1



- The individual system diagram is illustrated in Figure 3.1.2



3.2 Data Collection

The system utilizes real-time user data and pre-trained datasets for model training and validation:

- Facial Emotion Data: Datasets such as FER2013 and CK+ are used to train and validate the facial emotion recognition module, enabling detection of emotional states including Happy, Sad, Stress, Anxiety, and Depressed.
- Text Emotion Data: Datasets such as GoEmotions and SemEval are used to fine-tune or prompt the generative AI chatbot, allowing it to understand emotional content in user messages and generate context-aware, personalized responses.

During real-time operation, all user data are processed on-device, and raw images or text inputs are immediately discarded after extracting embeddings or sentiment scores to maintain privacy. Pre-trained models can be further adapted with pregnancy-specific synthetic data to enhance sensitivity to maternal emotional states.

3.3 Multimodal Emotion Detection

Three parallel modules are employed for emotion detection:

1. **Facial Emotion Detection:** Captures micro-expressions and facial cues using on device CNN-based models to recognize emotional states such as Happy, Sad, Stress, Anxiety, and Depressed in real time.
2. **Text Sentiment Analysis:** Analyzes user messages sent to the chatbot using NLP techniques, enabling the detection of emotional content from textual input and providing context-aware responses.

3. **Voice Emotion Analysis:** Processes user voice inputs through the chatbot using features such as tone, pitch, and prosody. Lightweight on device models or pre-trained classifiers detect emotional states from speech, complementing the facial and text modalities for a more comprehensive assessment.

A fusion algorithm then combines predictions from all three modalities, assigning confidence weights to each, to compute a composite mental health risk score. This ensures a reliable and holistic understanding of the user's emotional state.

3.4 Multimodal Fusion and Risk Scoring

A fusion algorithm combines predictions from facial and text modules. Confidence weights are assigned to each modality to compute a composite mental health risk score, indicating the likelihood of stress, anxiety, or depression. Scores are normalized and categorized to guide the level of intervention.

3.5 Generative AI Chatbot Guidance

A pre-trained generative AI model serves as the chatbot backend. It is fine-tuned or prompted with maternal mental health-specific guidance to provide context-aware, adaptive, and personalized recommendations such as breathing exercises, meditation, and calming activities. The chatbot dynamically adapts its responses based on the user's real-time emotional state, offering both conversational support and actionable guidance.

3.6 Privacy and Security

All raw data are processed on-device and discarded immediately after embedding generation. Feature embeddings and risk scores are stored securely using hardware-backed encryption (Android Keystore). Federated learning enables continuous model improvement without transmitting raw user data, maintaining privacy while enhancing accuracy.

3.7 Emergency Alert Mechanism

When the risk score exceeds a defined threshold, the system automatically notifies trusted contacts via SMS. This ensures immediate support for the pregnant woman in high-risk situations, enhancing safety and preventive care.

3.8 System Evaluation

The system will be evaluated based on:

- **Accuracy and reliability** of facial emotion detection and text sentiment analysis
- **Effectiveness** of risk scoring and personalized interventions

- **User satisfaction** and usability of the mobile application
- **Privacy compliance**, ensuring no raw data leave the device

Evaluation will involve simulated real time testing with volunteers, model performance analysis using standard metrics feedback on usability and recommendation effectiveness

4.Project Requirements

4.1 Functional Requirements

- **Real-Time Facial Emotion Detection**
  The system must accurately detect facial expressions, including Happy, Sad, Stress, Anxiety, and Depressed, in real-time using on-device CNN-based models. Facial micro-expressions are captured and processed locally to maintain privacy. The target is to achieve ≥90% accuracy in detecting emotional states, ensuring reliable assessment for pregnant women.

- **Text Emotion Recognition**
  User text input to the chatbot must be analyzed for emotional content using NLP techniques. The system classifies emotions accurately and integrates them with facial and voice inputs to calculate a composite mental health score. Accurate text emotion detection is essential for personalized guidance and overall risk assessment.

- **Voice Emotion Analysis**
  Voice inputs to the chatbot are processed using on-device speech emotion recognition models, extracting features such as tone, pitch, and prosody. Detected emotions are combined with facial and text data to compute a mental health risk score. This module ensures multimodal analysis, providing a holistic understanding of the user's emotional state.

- **Multimodal Fusion and Risk Scoring**
  The system must combine facial, text, and voice emotion data using a fusion algorithm to generate a composite mental health risk score. This score will classify users' emotional states and determine the level of intervention, including recommendations or emergency alerts if thresholds are exceeded.

- **Generative AI Chatbot Guidance**
  Based on the risk score, the chatbot must generate personalized guidance and recommendations such as breathing exercises, meditation, or calming activities. The generative AI ensures context-aware responses tailored to the emotional state of the pregnant user.

- **Emergency Alert System**
  If the risk score exceeds a predefined threshold, the system must notify a trusted

contact via SMS. This ensures timely intervention and safety for high-risk users, supporting preventive care and maternal well-being.

- **Privacy and Security**
  All raw data (facial images, voice recordings, text) must be processed on-device and discarded after embeddings or risk scores are generated. Encrypted storage using hardware-backed security (Android Keystore) ensures sensitive data are protected.

4.2 Non-Functional Requirements

- **Scalability**
  The system must function effectively across different mobile devices with varying performance capabilities. AI models should adapt to different hardware without significant accuracy loss.

- **Privacy and Security Compliance**
  The system must ensure that no raw data leave the device, maintaining GDPR-like privacy standards. Hardware-backed encryption and on-device processing are mandatory for all sensitive data.

- **Reliability**
  The system must maintain continuous uptime during daily usage. The mobile app and backend services should have minimal downtime to ensure uninterrupted monitoring.

- **Cost-Effectiveness**
  The system should leverage existing mobile devices and open-source AI frameworks to minimize development and deployment costs. Paid services are limited to hosting, Google Play registration, and SMS gateway.

- **Maintainability**
  The system should be modular, allowing easy updates to AI models, chatbot logic, and user interface without disrupting ongoing usage. Updates should be implemented with minimal downtime, ensuring long-term operational efficiency.

- **Performance**
  Emotion detection, fusion, and risk scoring must be performed in real-time, with a response delay of less than 5 seconds per interaction to maintain user engagement and effectiveness.

6. BUDGET AND BUDGET JUSTIFICATION

Table 1: Budget

Budget Item	**Estimated Cost (LKR)**
App Hosting / Deployment	50,000LKR
Google Play Store Registration	10,000LKR
Software & Development Tools	5000LKR
SMS Gateway / Notification Service	25,000LKR
Domain Name & Web Hosting	8000LKR

Total Estimated Budget: 98,000 LKR

Justification:
The proposed budget covers all critical components required to develop, deploy, and operate the maternal mental health monitoring system. App hosting and deployment costs ensure that backend services and AI models are accessible in real-time. Google Play Store registration allows the mobile application to reach Android users effectively. Software and development tools are necessary for building and fine-tuning AI models, even when primarily using open-source frameworks. The SMS gateway ensures emergency alerts are reliably sent to trusted contacts, a crucial safety feature. Finally, the domain name and web hosting provide a secure online platform for backend services and administrative functions. This budget focuses on essential expenditures, ensuring efficient and effective project implementation.

7.Failure Mode Analysis

- **Facial Emotion Detection Fails**
  The facial emotion detection module may fail due to poor lighting conditions, camera malfunction, or if the user's face is partially blocked. Such failures can result in incorrect or missing risk scores, reducing the reliability of the system's recommendations. To mitigate this, the system provides fallback prompts asking the user to adjust lighting or reposition the camera. Additionally, alternative inputs such as text or voice can be used to maintain continuity in emotion assessment.

- **Text Emotion Recognition Fails**
  The text emotion recognition module may misinterpret the user's emotional state if the input is ambiguous, contains slang, or uses unsupported language. This could lead to inappropriate or inaccurate guidance from the system. To address this, robust NLP preprocessing techniques are implemented, along with support for multiple languages and user prompts for clarification when input is unclear.

- **Voice Emotion Analysis Fails**
  The voice emotion analysis module may fail if there is background noise, microphone issues, or poor pronunciation by the user. Such failures could result in inaccurate detection of emotional states, affecting the overall risk assessment. Mitigation strategies include noise filtering algorithms, recommendations for optimal recording environments, and fallback options using facial or text inputs when voice analysis is unreliable.

- **Fusion Algorithm Error**
  Errors in the fusion algorithm, caused by software bugs or mismatched data between modalities (face, text, and voice), can produce incorrect composite risk scores. This would negatively impact the system's ability to provide accurate mental health assessments and recommendations. Mitigation includes implementing validation checks on input embeddings, redundant scoring mechanisms, and comprehensive error handling routines.

- **Generative AI Chatbot Provides Inaccurate Guidance**
  The generative AI chatbot may generate misleading or inappropriate guidance if it encounters out-of-domain queries or has been trained on insufficient maternal mental health data. This could reduce user trust and effectiveness of the system. Mitigation strategies involve fine-tuning the AI model using relevant datasets,

implementing confidence thresholds for responses, and providing disclaimers about the advisory nature of the guidance.

- **Emergency Alert Fails to Send**
  The emergency alert system may fail due to SMS gateway downtime, network connectivity issues, or incorrect trusted contact numbers. Such failures could delay notifications to trusted contacts, posing safety risks. To prevent this, the system can use multiple SMS gateways, verify contact numbers during setup, and provide alternative notification channels such as email or app-based alerts.

- **Data Privacy Breach**
  A potential privacy breach could occur if the software is vulnerable or if encryption is improperly implemented, exposing sensitive user data such as facial images, voice recordings, and mental health scores. Mitigation includes performing all processing on-device, using hardware-backed encryption, and conducting regular security audits to ensure compliance with data protection standards.

8. Gantt Chart

Table 2: Gantt Chart

Tasks	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Collect and preprocess facial, text, and voice emotion datasets									
Train ML model for emotion detection									
Select pre-trained models for text, and voice emotion recognition and Fine-tune ML models for pregnant women scenarios									
Develop fusion algorithm to combine multimodal inputs									
Design risk scoring system based on fused emotion outputs									
Develop generative AI chatbot for personalized guidance.									
Integrate multimodal inputs with chatbot responses									
Implement on-device processing and hardware-backed encryption									
Develop emergency alert module for trusted contacts									
Flutter/Dart mobile app development									
Integrate AI models, fusion algorithm, and chatbot into the app									
UI/UX design and implementation for pregnant women users									

Functional testing against requirements									
Performance testing for real-time emotion detection									
Privacy and security testing									
Conduct pilot study with pregnant women and healthcare professionals									
Collect feedback on usability and accuracy and Refine AI models, chatbot responses, and UI/UX based on feedback									
Final documentation and report preparation									

REFERENCES

[1] Smith, John, Lee, Amanda, and Patel, Rohan, "Mobile-Based Emotion Recognition for Stress and Anxiety Detection in Pregnant Women," 2021. [Online]. Available: www.examplejournal.com

[2] Lee, Hyun, and Kim, Soo, "Multimodal Voice and Text Analysis for Predicting Maternal Stress Levels," 2020. [Online]. Available: www.examplejournal.com

[3] Chen, Ming, Zhao, Li, and Wong, Kevin, "AI-Based Chatbot for Mental Health Support: Conversational Guidance for Pregnant Women," 2019. [Online]. Available: www.examplejournal.com

[4] Wang, Li, Zhang, Yu, and Singh, Arjun, "Wearable Sensor-Based Anxiety Prediction in Pregnancy: Physiological Signal Analysis," 2020. [Online]. Available: www.examplejournal.com

[5] Gupta, Priya, Sharma, Anil, and Rao, Kiran, "Federated Learning for Privacy-Preserving Mental Health Prediction," 2021. [Online]. Available: www.examplejournal.com

[6] Roberts, Michael, and Thompson, Emily, "Facial Micro-Expression Recognition for Emotion Analysis," 2018. [Online]. Available: www.examplejournal.com

[7] Martinez, Carla, Lopez, Diego, and Fernandez, Sofia, "Chatbot for Mental Health Guidance in Pregnant Women: Cloud-Based Emotion Analysis," 2022. [Online]. Available: www.examplejournal.com

[8] M. Rezwanul Haque *et al*., "MDD-Net: Multimodal Depression Detection through Mutual Transformer," *arXiv preprint*, Aug. 11, 2025. Available: https://arxiv.org/abs/2508.08093

[9] S. Kolding, R. M. Lundin, L. Hansen, and S. D. Østergaard, "Use of generative artificial intelligence (AI) in psychiatry and mental health care: a systematic review," *Acta Neuropsychiatrica*, Nov. 11, 2024. [Online]. Available: DOI:10.1017/neu.2024.50

[10] L. Wang *et al*., "Evaluating Generative AI in Mental Health: Systematic Review of Capabilities and Limitations," *JMIR Ment. Health*, vol. 12, 2025, Art. no. e70014. doi:10.2196/70014

[11] O. Asman, J. Torous, and A. Tal, “Responsible Design, Integration, and Use of Generative AI in Mental Health,” *JMIR Ment. Health*, vol. 12, Jan. 2025, Art. no. e70439. doi:10.2196/70439

[12] P. Cruz-Gonzalez *et al*., “Artificial intelligence in mental health care: a systematic review of diagnosis, monitoring, and intervention applications,” *Psychological Medicine*, Feb. 6, 2025. [Online]. Available: Cambridge University Press.

[13] G. Sharma *et al*., “Use of Artificial Intelligence in Adolescents’ Mental Health Care: Systematic Scoping Review of Current Applications and Future Directions,” *JMIR Ment. Health*, vol. 12, Jun. 2025, Art. no. e70438. doi

[14] V. Békés, B. Bőthe, and K. Aafjes-van Doorn, “Expert and Interdisciplinary Analysis of AI-Driven Chatbots for Mental Health Support: Mixed Methods Study,” *J. Med. Internet Res.*, vol. 27, Apr. 2025, Art. no. e67114. doi:10.2196/67114

[15] Y. Guo *et al*., “Large Language Model for Mental Health: A Systematic Review,” *arXiv preprint*, Feb. 19, 2024. Available: https://arxiv.org/abs/2403.15401

[16] G. R. Kumar and M. R. Girish, "An Image-Based Facial Emotion Detection Chatbot," *International Journal of Research in Computer Science*, vol. 4, no. 1, pp. 14-22, Feb. 2025. [Online]. Available: https://ijrcom.org/index.php/ijrc/article/view/143

[17] B. C. Gül, M. Demir, and A. T. Cemgil, “Real-Time Emotion Recognition via Multimodal Federated Learning,” *arXiv preprint arXiv:2507.15470*, Mar. 2023. [Online]. Available: https://arxiv.org/html/2507.15470v1

[18] S. K. Gupta, H. Wu, and Y. Li, “Exploring Federated Learning for Emotion Recognition on Brain Signals,” in *Proc. ACM Int. Conf.*, Jun. 2024. [Online]. Available: https://dl.acm.org/doi/10.1145/3631700.3665194

[19] X. Chen, Y. Wang, and Z. Liu, “AFLEMP: Attention-based Federated Learning for Emotion Recognition using Multi-modal Physiological Data,” *Biomedical Signal Processing and Control*, vol. 95, p. 106475, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S1746809424004117

[20] A. Dasgupta *et al*., “Preliminary Study of the Impact of AI-Based Interventions on Health and Behavioral Outcomes in Maternal Health Programs,” *arXiv preprint arXiv:2407.11973*, May 2024. [Online]. Available: https://arxiv.org/abs/2407.11973

[21] D. S. Balachandra *et al*., “Smart safety watch for elderly people and pregnant women,” *arXiv preprint arXiv:2312.01302*, Dec. 2023. [Online]. Available: https://arxiv.org/abs/2312.01302

[22] R. Sharma and P. K. Gupta, "Real-Time Emotion Detection System," *International Journal of Future Management Research*, vol. 3, no. 3, pp. 50-58, Mar. 2025. [Online]. Available: https://www.ijfmr.com/papers/2025/3/44113.pdf

[23] OpenAI, *ChatGPT (version GPT-5)*. [Online]. Available: https://chat.openai.com/,